Deepika Paidepelly | Data Engineer

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PROFESSIONAL SUMMARY:

- Over 10+ years of extensive experience as a Sr. Data Engineer, specializing in cloud platforms, big data technologies, data processing, and data integration.
- Proficient in working with AWS (EC2, S3, RDS, Lambda, Glue, DynamoDB, Kinesis) and Azure (Data Factory, Databricks, CosmosDB) services to build scalable, efficient data solutions.
- Strong expertise in Big Data technologies, including Apache Spark, Hadoop, Hive, Kafka, and Flink for designing and implementing data pipelines.
- Deep understanding of Data Engineering concepts such as ETL, data warehousing, star schema design, and data lineage to ensure high-quality data integration and availability.
- Adept at utilizing Python, SQL, and PowerShell to build and optimize data pipelines and models and automate tasks.
- Experienced in Machine Learning & Data Science, specifically in predictive modeling, NLP, and generative AI (Gen AI) to provide data-driven insights and decision-making tools.
- Hands-on experience in DevOps & Automation tools such as Docker, Kubernetes, Terraform, and Jenkins to streamline the deployment and monitoring of data systems.
- Knowledgeable in Data Visualization tools such as Power BI, Tableau, and Matplotlib, and ensured complex data insights are easily accessible and actionable for stakeholders.
- Solid understanding of Data Governance, real-time streaming, encryption, and RBAC to ensure data security, compliance, and scalability.
- Strong collaboration and communication skills for cross-functional alignment.
- Proactive problem-solving, ensuring on-time, on-budget project delivery.
- Rapidly adapts to new technologies, driving process and system improvements.

TECHNICAL SKILLS:

Cloud Platforms & Services: AWS (EC2, S3, RDS, Lambda, Glue, EMR, DynamoDB, Kinesis, Data Pipeline, CloudTrail, CloudWatch, IAM, KMS, CloudFormation), Azure (Data Factory, Databricks, Functions, Logic Apps, Active Directory, Snowflake, CosmosDB, DevOps)
Big Data Technologies: Apache Spark, Apache Hadoop, Apache Hive, Apache Kafka, Apache Flink, Apache Airflow, Apache Sqoop, Apache Beam, HDFS, Databricks, Matillion
Data Processing & Analysis: Pandas, NumPy, SciPy, D3.js, Seaborn
Machine Learning & Data Science: Generative AI (Gen AI), Predictive Modeling, NLP (spaCy)
ETL, Data Integration & Warehousing: Informatica, Snowflake, PL/SQL, Star Schema, ETL Workflows, Data Lineage
Programming & Scripting: Python, SQL, PowerShell, VBA
Agile & Project Management Tools: Jira, Confluence, Trello, Scrum, Kanban
DevOps & Automation: Docker, Kubernetes, Jenkins, Terraform, Git, GitHub
Databases: MySQL, PostgreSQL, MongoDB, SQL Server
Data Visualization & Reporting: Power BI, Tableau, Lucidchart, Excel (VLOOKUPs, Pivot Tables, VBA, Macros)
Concepts & Techniques: Real-Time Streaming, Encryption, Metadata, Data Governance, RBAC, ML Models, Forecasting, Anomaly Detection

EXPERIENCE:

SPECTRUM, Stamford, CT
Sr. Data Engineer | March 2024 – Present
Roles & Responsibilities:

- Automated ETL workflows by designing and implementing AWS Step Functions, streamlining data processing across various AWS services.
- Created and maintained scalable RESTful APIs to facilitate secure data exchange between distributed applications and third-party services.
- Designed and implemented end-to-end ETL workflows using Informatica, ensuring smooth data migration across multiple environments while maintaining high data integrity and governance standards.
- Developed and optimized PL/SQL queries and stored procedures to enhance metadata accuracy, improve database indexing, and boost performance in Amazon Athena.
  Built and deployed real-time data streaming solutions with Kafka and Kafka Streams, enabling efficient handling and transformation of large-scale datasets.

- Managed and optimized data lakes by integrating AWS Lake Formation, ensuring robust encryption and controlled access through AWS IAM policies.
- Hadoop and Apache Spark were used to process large datasets, enhancing distributed computing efficiency for complex analytical workloads.
- Created and optimized Matillion ETL workflows for seamless data transformation and loading into Snowflake, improving overall data governance and operational efficiency.
- Designed and optimized Snowflake schemas to support high-performance data warehousing, improving reporting and analytics capabilities.
- Developed CI/CD pipelines with Jenkins and Terraform to automate infrastructure provisioning, streamline code integration, and reduce deployment times.
- Configured AWS Lambda functions for serverless data processing, improving efficiency and reducing operational costs in cloud-based ETL workflows.
- Built and managed Databricks clusters, leveraging Apache Spark to enable high-performance distributed computing and maintain comprehensive data lineage tracking.
- Deployed containerized data processing workflows using Docker and Kubernetes, ensuring high availability and seamless scalability on Amazon EC2 and Amazon EMR.
- Designed and implemented metadata management solutions, improving data asset visibility, discovery, and usability across the organization.
- Engineered real-time data streaming pipelines with Kafka and enabled seamless integration between multiple enterprise systems.
- Implemented advanced encryption techniques to secure sensitive data stored in Amazon S3, Snowflake, and other cloud storage solutions.
- Automated data migration workflows to ensure a smooth transition between on-premises and AWS cloud environments with minimal downtime.
- Created data visualizations using Python and Matplotlib to analyze trends, generate insights, and support business intelligence initiatives.
- Applied advanced indexing strategies to optimize query performance in Amazon Athena, Snowflake, and other relational database systems.
- Developed and maintained data lineage tracking mechanisms, enhancing traceability across ETL workflows for audit and compliance purposes.
- Utilized AWS EMR to execute large-scale Hadoop and Apache Spark workloads, efficiently processing structured and unstructured data.
- Designed and implemented robust data validation frameworks, applying business logic to maintain data accuracy, compliance, and integrity.
- Established business intelligence (BI) frameworks by developing interactive dashboards and reporting solutions to drive data-driven decision-making.
- Developed and managed data governance for compliance and best practices.

Environment: AWS Step Functions, Amazon EC2, Amazon EMR, Amazon S3, Hadoop, Apache Spark, Matillion, RESTful APIs, Informatica, PL/SQL, Amazon Athena, Kafka, Kafka Streams, AWS Lake Formation, AWS IAM, Snowflake, Jenkins, Terraform, AWS Lambda, Databricks, Docker, Kubernetes, Python, Matplotlib.

CALIBER TECHNOLOGIES PVT. LTD, India
Data Engineer | July 2020 – December 2023
Roles & Responsibilities:

- Developed and deployed Azure Functions to support event-driven data processing, reducing latency and enhancing real-time analytics capabilities.
- Streamlined job orchestration and workflow automation with Apache Airflow and improved pipeline reliability and operational efficiency.
- Engineered robust data pipelines in Databricks, optimizing Spark workloads for enhanced performance and scalability in large-scale analytics.
- Architected and implemented high-performance data processing solutions using Apache Spark and PySpark to efficiently handle large-scale structured and unstructured datasets.
- Designed and optimized ETL workflows using Azure Data Factory (ADF), automating data ingestion, transformation, and integration from various sources to improve processing speed and accuracy.
- Implemented and maintained Azure Synapse Analytics solutions to enable interactive query execution and efficient big data processing.
- Designed and deployed Azure Data Lake Storage (ADLS) solutions to create a secure, scalable data lake architecture that supports advanced analytics.

- Developed and optimized Azure SQL Database schemas, improving query performance through indexing strategies and efficient data modeling.
- Created and maintained star schema models to enhance analytical query performance and support business intelligence reporting.
- Built Azure Stream Analytics pipelines to process real-time data streams, supporting anomaly detection and predictive analytics initiatives.
- Leveraged PowerShell scripting to automate infrastructure provisioning, optimize Azure resource management, and enhance DevOps workflows.
- Designed and managed scalable, high-performance NoSQL databases, including MongoDB, to support flexible data storage and retrieval needs.
- Developed and deployed machine learning models to enhance data-driven decision-making processes and uncover actionable business insights.
- Designed advanced forecasting models to predict trends and optimize strategic decision-making through data-driven analytics.
- Implemented anomaly detection techniques to ensure data quality and identified inconsistencies, and maintained data integrity across systems.
- Integrated Azure DevOps and Bitbucket for seamless version control, CI/CD pipeline automation, and efficient deployment workflows.
- Designed and implemented end-to-end CI/CD pipelines for data engineering projects and enabled rapid, reliable, and scalable deployments.
- Established and enforced data governance best practices to ensure regulatory compliance, security, and effective data lifecycle management.
- Monitored and optimized data pipelines using Azure Monitor and Application Insights and ensured high availability and system reliability.
- Designed interactive Power BI dashboards and reports to give stakeholders actionable insights and real-time analytics.
- Configured and managed role-based access control (RBAC) in Azure to enforce security policies and safeguard sensitive data.

Environment: Azure Functions, Apache Airflow, Databricks, Apache Spark, PySpark, ADF, Azure Synapse Analytics, ADLS, Azure SQL Database, Azure Stream Analytics, PowerShell, MongoDB, NoSQL, Azure DevOps, Bitbucket, Azure Monitor, Application Insights, Power BI, RBAC.

Careator Technologies Pvt Ltd., India
Data Engineer | January 2018 - June 2020
Roles & Responsibilities:

- Worked successfully in a team environment using Agile Scrum methodologies.
- Implemented code in Python to retrieve and manipulate data.
- Developed Python and Shell Scripts for automation of the build and release process.
- Used Pandas as API to put the data as time series and tabular format for manipulation and retrieval of data.
- Developed Restful service using Python Flask framework.
- Implemented PySpark jobs for Batch Analysis.
- Developed a data platform from scratch and took part in requirement gathering and analysis phase of the project in documenting the business requirements.
- Assisted with tasks such as data pipeline engineering, data analytics, data scrapping and mining, data visualization, data dashboarding, artificial intelligence (AI), machine learning (ML), AWS Cloud Computing.
- Helped in the development of the Risk Scoring for supply chain procurement along with KPIs and MVP (Minimum Viable Product).
- Implemented AWS Lambdas to drive real-time monitoring dashboards from system logs.
- Monstered Cloud watch dashboard for traffic inbound in the AWS environment.
- Installed the application on AWS EC2 instances and configured the storage on S3 buckets.
- Analysed the data to report patterns and trends that enabled business teams to make informed Structured and led customer feedback analysis projects in error detection Using Python libraries.
- Interpreted problems and provided solutions to business problems using data analysis, data mining, optimization tools, and machine learning techniques and statistics.
- Analysed SQL scripts and designed the solutions to implement using PySpark.
- Worked on data pre-processing and cleaning the data to perform feature engineering and performed data imputation techniques for the missing values in the dataset using Python.
- Created various types of data visualizations using Python and Tableau.

- Generated the reports of the results of the scripts to analyse the necessities by using data visualization toll Tableau.
- Implemented row level security on data and have an understanding of application security layer models in Power BI.
- Wrote, compiled, and executed programs as necessary using Apache Spark in Scala to perform ETL jobs with ingested data.
- Worked on development of SQL and stored procedures on MySQL.
- Designed and developed a horizontally scalable APIs using Python Flask.
- Handled the day to day issues and fine tuning the applications for enhanced performance.

Environment: Python, Flask, Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn, PySpark, Spark, Tableau, MySQL, PowerBI, AWS EC2, Lambdas, S3, SQL, Linux, Windows

FOCO IT CONSULTING SERVICES PVT. LTD, India
Data Analyst | June 2014 – December 2017
Roles & Responsibilities:

- Designed dynamic dashboards in Excel (VLOOKUPs, Pivot Tables, VBA, and Macros) and Google Sheets to enhance data visualization and reporting.
- Led data validation and business intelligence efforts to maintain data integrity and enable accurate, data-driven decision-making.
- Managed scalable AWS data storage solutions, ensuring optimal performance, security, and compliance with industry standards.
- Automated and streamlined ETL workflows, enhancing data pipeline efficiency and reducing processing time.
- Optimized SQL and Oracle database queries to enhance data retrieval, performance tuning, and business intelligence reporting.
- Utilized Python for data cleansing, transformation, and automation, and improved data accuracy and streamlining market research processes.
- Extracted industry and competitor insights through web scraping with Scrapy and supported market analysis and strategic decision-making.
- Strengthened data security by implementing encryption, access controls, and compliance measures to protect sensitive business information.
- Developed automated reporting systems with real-time data integration, utilizing SAP BusinessObjects, Google Sheets, and Excel for enhanced business intelligence.
- Defined and monitored key performance indicators (KPIs) to measure operational success and guide strategic decision-making.
- Created Lucidchart workflows and data models to optimize business operations and enhance process efficiencies.
- Improved task allocation, progress tracking, and team collaboration using Trello, increasing productivity and workflow efficiency.
- Utilized Git for version control, ensuring structured collaboration and seamless tracking of changes in data-driven projects.
- Delivered actionable insights on competitor trends, industry performance, and operational metrics to inform business strategy.

Environment: Excel (VLOOKUPs, Pivot Tables, VBA, Macros), Google Sheets, AWS, SQL, Oracle, Python, Scrapy, SAP BusinessObjects, Lucidchart, Trello, Git.

EDUCATION:

Bachelor of Technology in Computer Science, JNTUH India - 2014